

		the reading of X increases.
24	D	For a long straight wire, $B = \frac{\mu_0 I}{2\pi d}$ At Q, $B_Q = B_{\text{earth}} - (B_{\text{due to P}} + B_{\text{due to R}})$ $= (2.0 \times 10^{-5}) - 2 \left(\frac{(4\pi \times 10^{-7})(1.0)}{2\pi(0.30)} \right)$ $= 1.8667 \times 10^{-5} \text{ T (northwards)}$ $\frac{F}{L} = B_Q I_Q = (1.8667 \times 10^{-5})(2.0)$ $= 3.7334 \times 10^{-5} = 3.7 \times 10^{-5} \text{ N m}^{-1}$ Using Fleming's Left Hand Rule, the direction of the force is towards the West.
25	A	In field Y, the magnetic forces on the sides QR and RS of the coil are constant in